

successfully into larger and larger markets until it became the world's largest automaker in 2008.

The biggest advantage of working in small batches is that quality problems can be identified much sooner. This is the origin of Toyota's famous andon cord, which allows any worker to ask for help as soon as they notice any problem, such as a defect in a physical part, stopping the entire production line if it cannot be corrected immediately. This is another very counterintuitive practice. An assembly line works best when it is functioning smoothly, rolling car after car off the end of the line. The andon cord can interrupt this careful flow as the line is halted repeatedly. However, the benefits of finding and fixing problems faster outweigh this cost. This process of continuously driving out defects has been a win-win for Toyota and its customers. It is the root cause of Toyota's historic high quality ratings and low costs.

SMALL BATCHES IN ENTREPRENEURSHIP

When I teach entrepreneurs this method, I often begin with stories about manufacturing. Before long, I can see the questioning looks: what does this have to do with my startup? The theory that is the foundation of Toyota's success can be used to dramatically improve the speed at which startups find validated learning.

Toyota discovered that small batches made their factories more efficient. In contrast, in the Lean Startup the goal is not to produce more stuff efficiently. It is to—as quickly as possible—learn how to build a sustainable business.

Think back to the example of envelope stuffing. What if it turns out that the customer doesn't want the product we're building? Although this is never good news for an entrepreneur, finding out sooner is much better than finding out later. Working in small batches ensures that a startup can minimize the expenditure of time, money, and effort that ultimately turns out to have been wasted.